Unit 9: File Handling

PHP File Handling

PHP File System allows us to create file, read file line by line, read file character by character, write file, append file, delete file and close file.

1.PHP Open File - fopen()

The PHP fopen() function is used to open a file.

Syntax

resource fopen (string $filename , string $mode [, bool $use_include_path = false [, resource $context]])

The file may be opened in one of the following modes:

Modes	Description
r	**Open a file for read only**. File pointer starts at the beginning of the file
w	**Open a file for write only**. Erases the contents of the file or creates a new file if it doesn't exist. File pointer starts at the beginning of the file
a	**Open a file for write only**. The existing data in file is preserved. File pointer starts at the end of the file. Creates a new file if the file doesn't exist
x	**Creates a new file for write only**. Returns FALSE and an error if file already exists
r+	**Open a file for read/write**. File pointer starts at the beginning of the file

w+	**Open a file for read/write**. Erases the contents of the file or creates a new file if it doesn't exist. File pointer starts at the beginning of the file
a+	**Open a file for read/write**. The existing data in file is preserved. File pointer starts at the end of the file. Creates a new file if the file doesn't exist
x+	**Creates a new file for read/write**. Returns FALSE and an error if file already exists

2.PHP Close File - fclose()

The PHP fclose() function is used to close an open file pointer.

Syntax

bool fclose (resource $handle)

```
<?php
fclose($handle);
?>
```

3.PHP Read File - fread()

The PHP fread() function is used to read the content of the file. It accepts two arguments: resource and file size.

Syntax

string fread (resource $handle , int $length)

Example

```
<?php
$filename = "c:\\myfile.txt";
$handle = fopen($filename, "r");//open file in read mode
$contents = fread($handle, filesize($filename));//read file
echo $contents;//printing data of file
fclose($handle);//close file
?>
```

Output

```
hello php file
```

Example

```
<!DOCTYPE html>
<html>
<body>
<?php
$myfile = fopen("webdictionary.txt", "r")
or die("Unable to open file!");
echo
fread($myfile,filesize("webdictionary.txt")
);
fclose($myfile);
?>
</body>
</html>
```

4.PHP Write File - fwrite()

The PHP fwrite() function is used to write content of the string into file.

Syntax

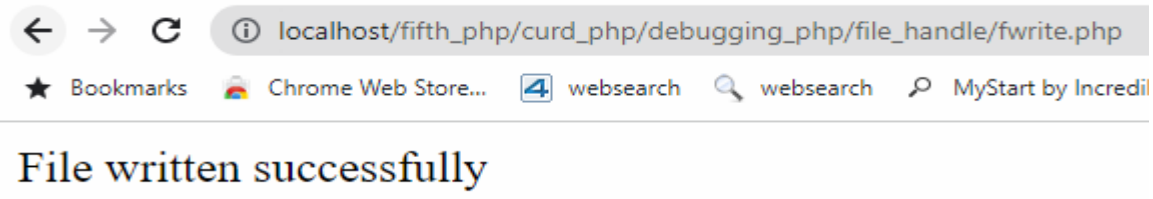
int fwrite (resource $handle , string $string [, int $length])

Example

```
<?php
```

```
$fp = fopen('data.txt', 'w');//open file in
write mode
fwrite($fp, 'hello ');
fwrite($fp, 'subash');
fclose($fp);

echo "File written successfully";
?>
```

Output



File written successfully

5.PHP Append to File - fwrite()

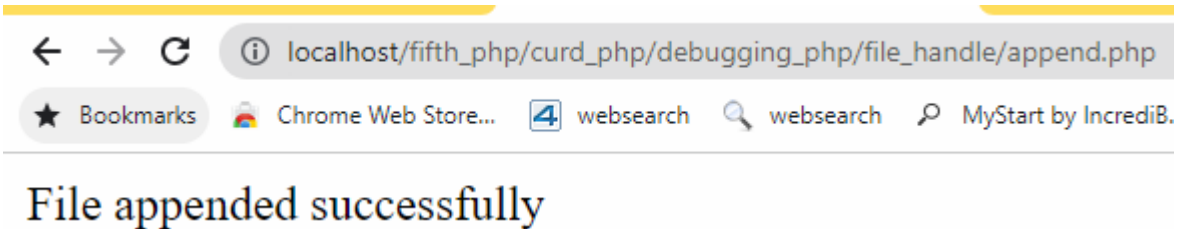
The PHP fwrite() function is used to write and append data into file.

Example

```
<?php
$fp = fopen('data.txt', 'a');//opens file
in append mode
fwrite($fp, ' this is additional text ');
fwrite($fp, 'appending data');
fclose($fp);
```

```
echo "File appended successfully";
?>
```

Output: data.txt



6.PHP Delete File - unlink()

The PHP unlink() function is used to delete file.

Syntax

bool unlink (string $filename [, resource $context])

Example

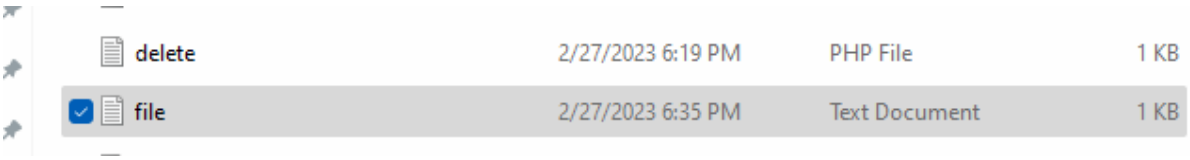
<?php

unlink('data.txt');

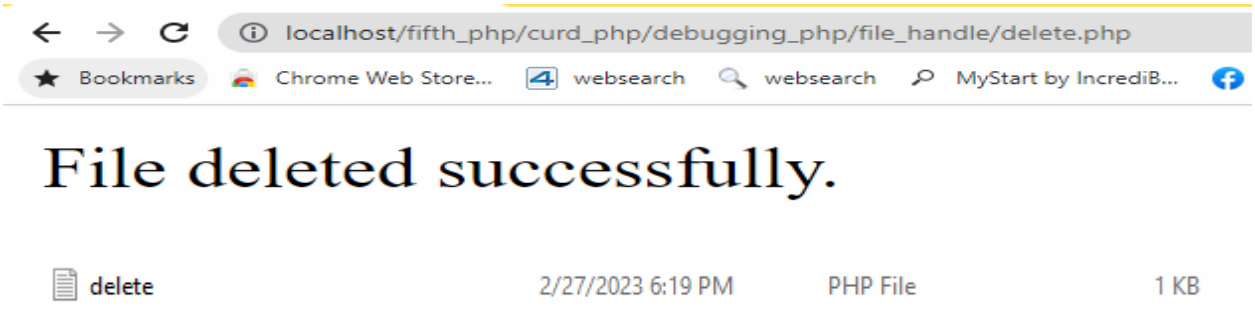
echo "File deleted successfully";

?>

```
<?php
$file_path = "file.txt";

if (unlink($file_path)) {
  echo "File deleted successfully.";
} else {
  echo "Error deleting file.";
}
?>
```



After



PHP File Upload

PHP allows you to upload single and multiple files through few lines of code only.

- PHP file upload features allows you to upload binary and text files both. Moreover, you can have the full control over the file to be uploaded through PHP authentication and file operation functions.

PHP $_FILES

The PHP global $_FILES contains all the information of file. By the help of $_FILES global, we can get file name, file type, file size, temp file name and errors associated with file.

Here, we are assuming that file name is *filename*.

$_FILES['filename']['name']

returns file name.

$_FILES['filename']['type']

returns MIME type of the file.

$_FILES['filename']['size']

returns size of the file (in bytes).

$_FILES['filename']['tmp_name']

returns temporary file name of the file which was stored on the server.

$_FILES['filename']['error']

returns error code associated with this file.

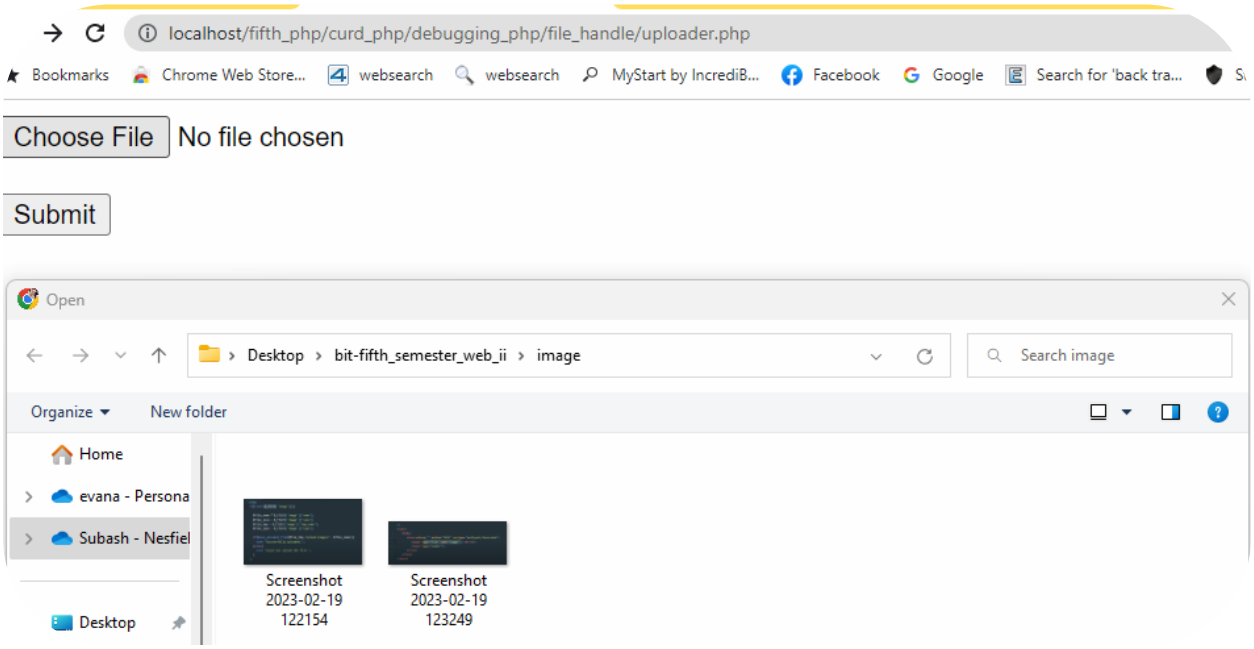
move_uploaded_file() function

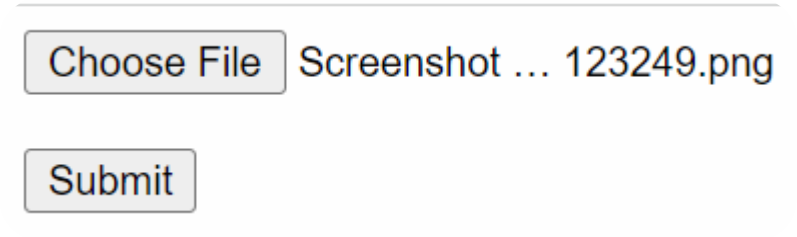
The move_uploaded_file() function moves the uploaded file to a new location. The move_uploaded_file() function checks internally if the file is uploaded thorough the POST request. It moves the file if it is uploaded through the POST request.

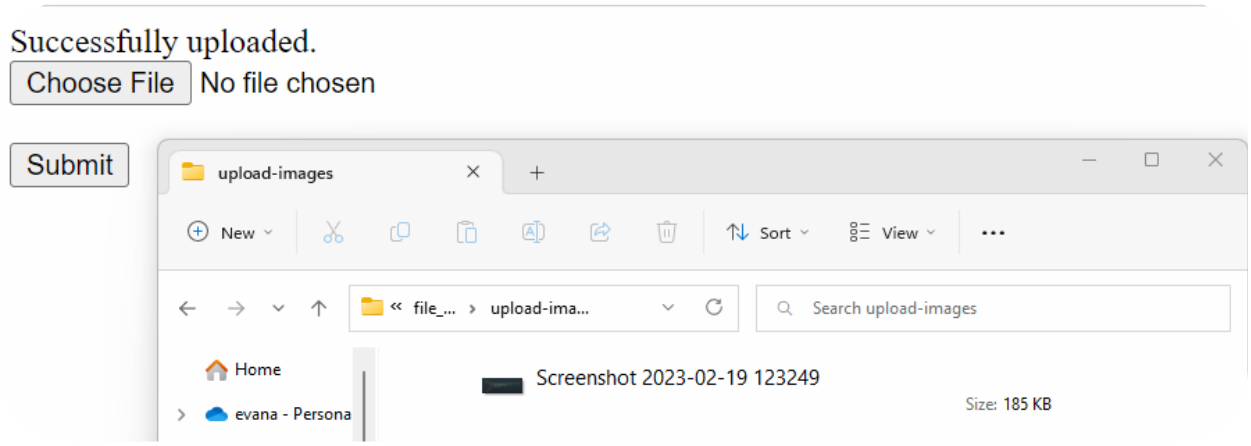
Syntax:bool move_uploaded_file (string $filename , string $destination)

```
<?php
if (isset($_FILES['image']))
{
$file_name = $_FILES['image']['name'];
$file_size = $_FILES['image']['size'];
$file_tmp = $_FILES['image']['tmp_name'];
$file_type = $_FILES['image'][ 'type'];
if(move_uploaded_file($file_tmp, "upload-
images/". $file_name))
{
echo "Successfully uploaded.";
}
else
{
echo "Could not upload the file.";
}
```

```
}
?>
<html>
<body>
<form action="" method="POST"
enctype="multipart/form-data">
<input type="file" name="image"/><br><br>
<input type="submit"/>
</form>
</body>
</html>
```







Parsing CSV File

csv file

A CSV (Comma Separated Values) file is a simple text file that stores tabular data in a plain text format, where each line represents a row, and each field within the row is separated by a comma (or sometimes a different delimiter character, such as a semicolon or a tab).

Here's an example of what a CSV file might look like:

Name, Age, City

John Doe, 30, New York

Jane Smith, 25, London

Bob Johnson, 40, San Francisco

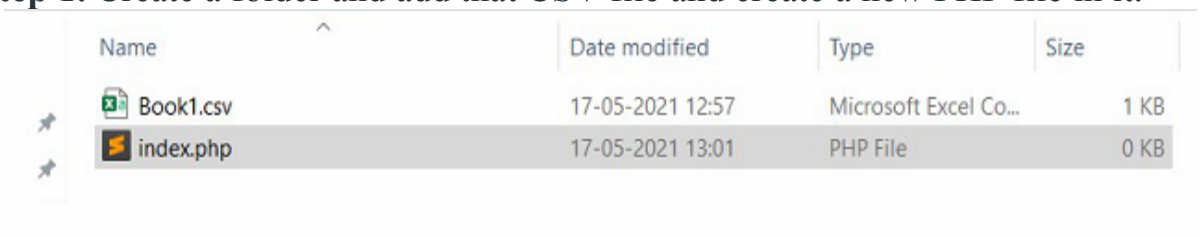
In this example, the first row represents the column headers, and each subsequent row represents a record. The data for each record is

separated by commas, with each field representing a different attribute of the record (in this case, name, age, and city).

CSV files are a popular way to exchange data between different applications or systems, as they can be easily read and written by many different programs and programming languages. In PHP, for example, you can use the fgetcsv() function to read data from a CSV file, and the fputcsv() function to write data to a CSV file.

To parse a CSV file in PHP, you can use the built-in fgetcsv() function, which reads a line from a file pointer and parses it as CSV.

Step 1. Create a folder and add that CSV file and create a new PHP file in it.



Name	Date modified	Type	Size
Book1.csv	17-05-2021 12:57	Microsoft Excel Co...	1 KB
index.php	17-05-2021 13:01	PHP File	0 KB

file path

Step 2. Open the PHP file and write the following code in it which is explained in the following steps.

1. Open dataset of CSV using fopen function.
2. $open = fopen("filename.csv", "r");
3. Read a line using **fgetcsv()** function.
   $data = fgetcsv($Open, 1000, ",");
4. Use a loop to iterate in every row of data.

   ```
   while (($data = fgetcsv($Open, 1000, ",")) !== FALSE)
   {
     // Read the data
   }
   ```

   Close that file using PHP **fclose()** method.
5. fclose($open);
6. <?php

```
7.
8.      if (($open = fopen("Book2.csv",
  "r")) !== FALSE)
9.      {
10.
11.       while (($data = fgetcsv($open,
  1000, ",")) !== FALSE)
12.       {
13.         $array[] = $data;
14.       }
15.
16.       fclose($open);
17.     }
18.     echo "<pre>";
19.     //To display array data
20.     var_dump($array);
21.     echo "</pre>";
22.   ?>
```

localhost/fifth_php/curd_php/debugging_php/file_handle/parsecsv.p.

Bookmarks Chrome Web Store... websearch websearch MyStart by IncrediB...

```
array(4) {
  [0]=>
  array(2) {
    [0]=>
    string(3) "ram"
    [1]=>
    string(1) "1"
  }
  [1]=>
  array(2) {
    [0]=>
    string(4) "hari"
    [1]=>
    string(2) "22"
  }
  [2]=>
  array(2) {
    [0]=>
    string(4) "sita"
    [1]=>
    string(2) "45"
  }
  [3]=>
  array(2) {
    [0]=>
    string(4) "gita"
    [1]=>
    string(2) "12"
  }
```

JSON File

A JSON (JavaScript Object Notation) file is a lightweight data interchange format that is easy for humans to read and write, and easy for machines to parse and generate. It is based on a subset of the JavaScript programming language, and is often used to transmit data between a server and a web application, as well as between different systems and platforms.

JSON data is represented as key-value pairs, with values that can be arrays, objects, or primitive data types such as strings, numbers, and booleans. Here's an example of what a JSON file might look like:

```
{
    "name": "John Doe",
    "age": 30,
    "city": "New York",
    "pets": [
        {
            "name": "Fluffy",
            "type": "cat"
        },
        {
            "name": "Fido",
            "type": "dog"
        }
    ]
}
```

In this example, the JSON data represents a person named John Doe, with an age of 30 and a city of New York. The person has two pets, Fluffy the cat and Fido the dog, represented as objects within an array.

In PHP, you can use the built-in json_decode() function to parse a JSON file into a PHP variable, and the json_encode() function to encode a PHP variable as a JSON string. These functions make it easy to work with JSON data in your PHP applications.

1.Use **file_get_contents()** function to read JSON file into PHP. This function is used to read the file into PHP code.

Syntax:

file_get_contents(path, file_name)

- *file_name* is the name of the file and *path* is the location to be checked.

2.Use **json_decode()** function to decode to JSON file into array to display it. It is used to convert the JSON into an array.

Syntax:

json_decode($json_object, true)

- $json_object is the file object to be read.

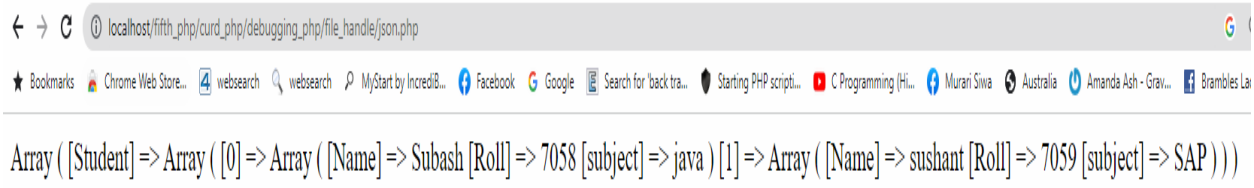
PHP code: The following is the PHP code to parse JSON file.

Example:

```
<?php

// Read the JSON file
$json = file_get_contents('my_data.json');

// Decode the JSON file
$json_data = json_decode($json,true);

// Display data
print_r($json_data);
```

```
?>
```

Output:



Array ([Student] => Array ([0] => Array ([Name] => Subash [Roll] => 7058 [subject] => java) [1] => Array ([Name] => sushant [Roll] => 7059 [subject] => SAP)))